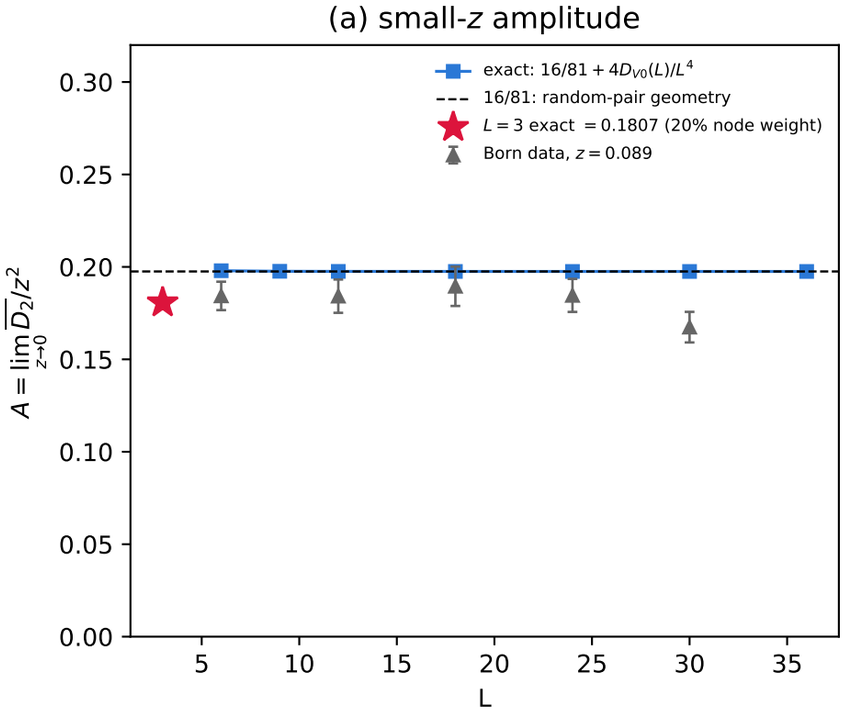
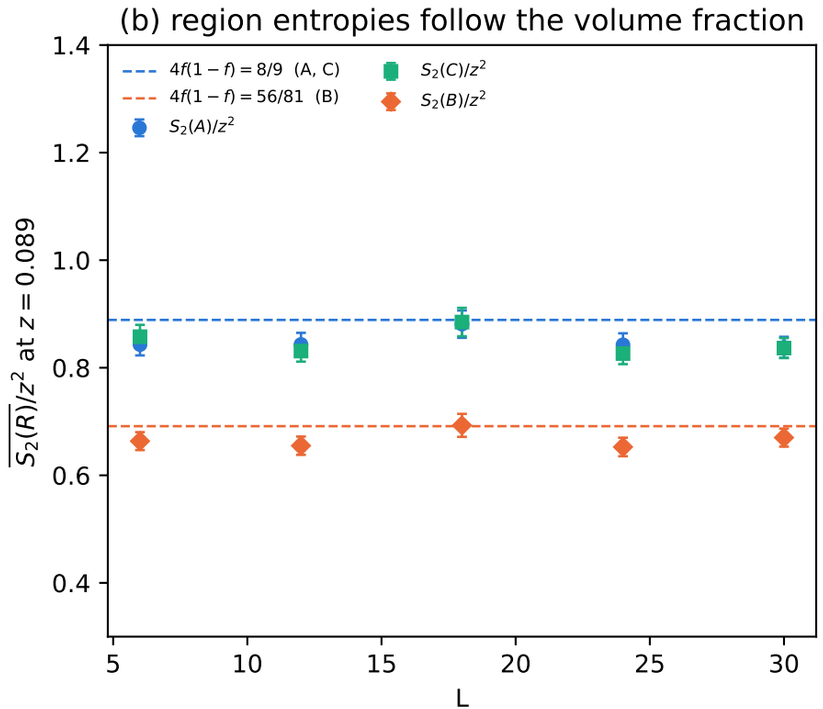
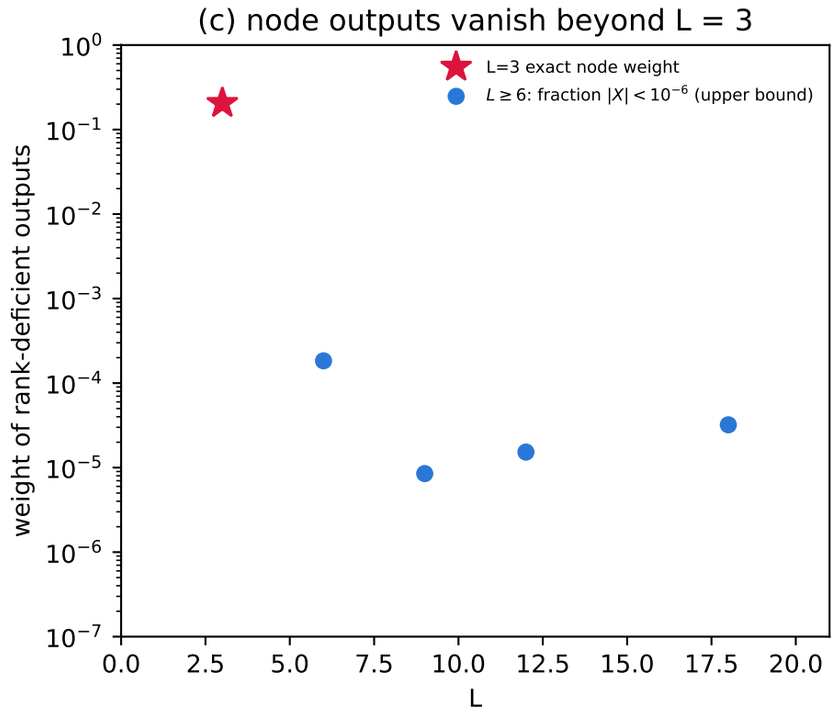

(a) small-z amplitude
exact: 16/81 + 4D_V0(L)/L^4
16/81: random-pair geometry
L = 3 exact = 0.1807 (20% node weight)
Born data, z = 0.089
A = lim_{z→0} D̄_2/z^2
L


(b) region entropies follow the volume fraction
4f(1 − f) = 8/9 (A, C)
4f(1 − f) = 56/81 (B)
S_2(A)/z^2
S_2(C)/z^2
S_2(B)/z^2
S_2(R)/z^2 at z = 0.089
L


(c) node outputs vanish beyond L = 3
L=3 exact node weight
L ≥ 6: fraction |X| < 10^−6 (upper bound)
weight of rank-deficient outputs
L